

CLASS ACTIVITY (07.08.2026)

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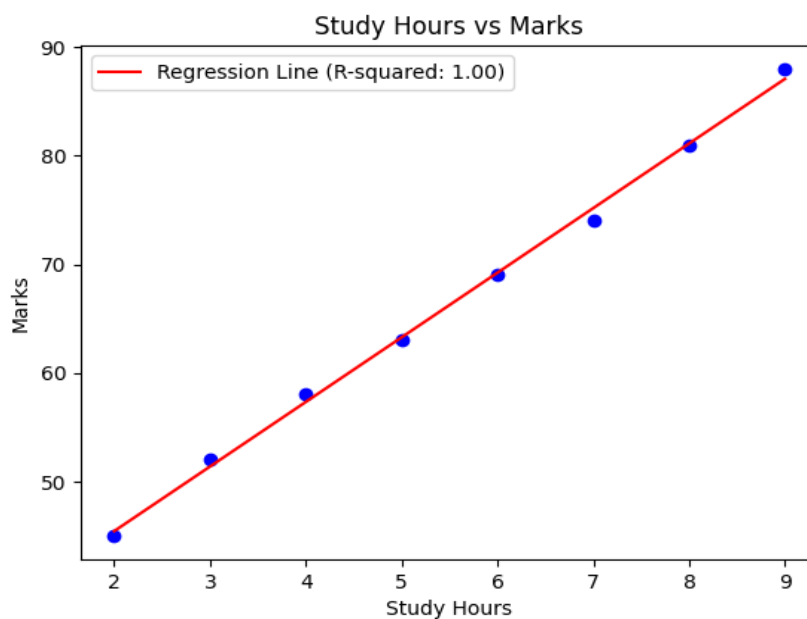
Reg No: 192424100

Q1. Scatter Plot – Study Hours vs Marks

Python Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
study = pd.DataFrame({
    'StudyHours': [2, 3, 4, 5, 6, 7, 8, 9],
    'Marks': [45, 52, 58, 63, 69, 74, 81, 88]
})
plt.scatter(study['StudyHours'], study['Marks'], color='blue')
plt.title('Study Hours vs Marks')
plt.xlabel('Study Hours')
plt.ylabel('Marks')
slope, intercept, r_value, p_value, std_err = stats.linregress(study['StudyHours'],
study['Marks'])
plt.plot(study['StudyHours'], intercept + slope * study['StudyHours'], color='red',
label=f'Regression Line (R-squared: {r_value**2:.2f})')
plt.legend()
plt.show()
correlation = study['StudyHours'].corr(study['Marks'])
print(f"Correlation between StudyHours and Marks: {correlation:.2f}")
```

Output:

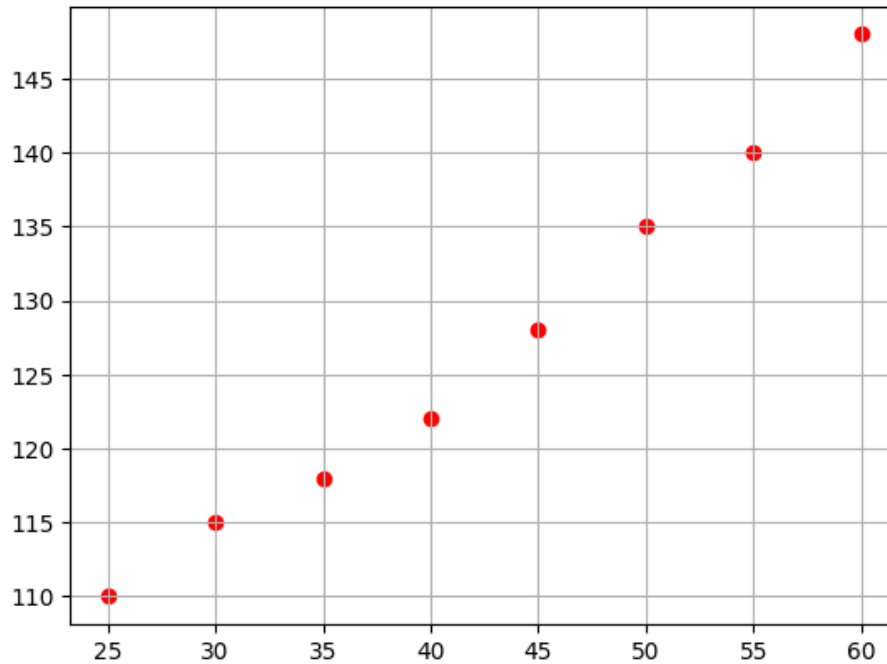


Q2. Scatter Plot – Age vs Systolic Blood Pressure

Python Code:

```
import matplotlib.pyplot as plt
age=[25,30,35,40,45,50,55,60]
bp=[110,115,118,122,128,135,140,148]
plt.scatter(age,bp,color='red')
plt.grid(True)
plt.show()
```

Output:

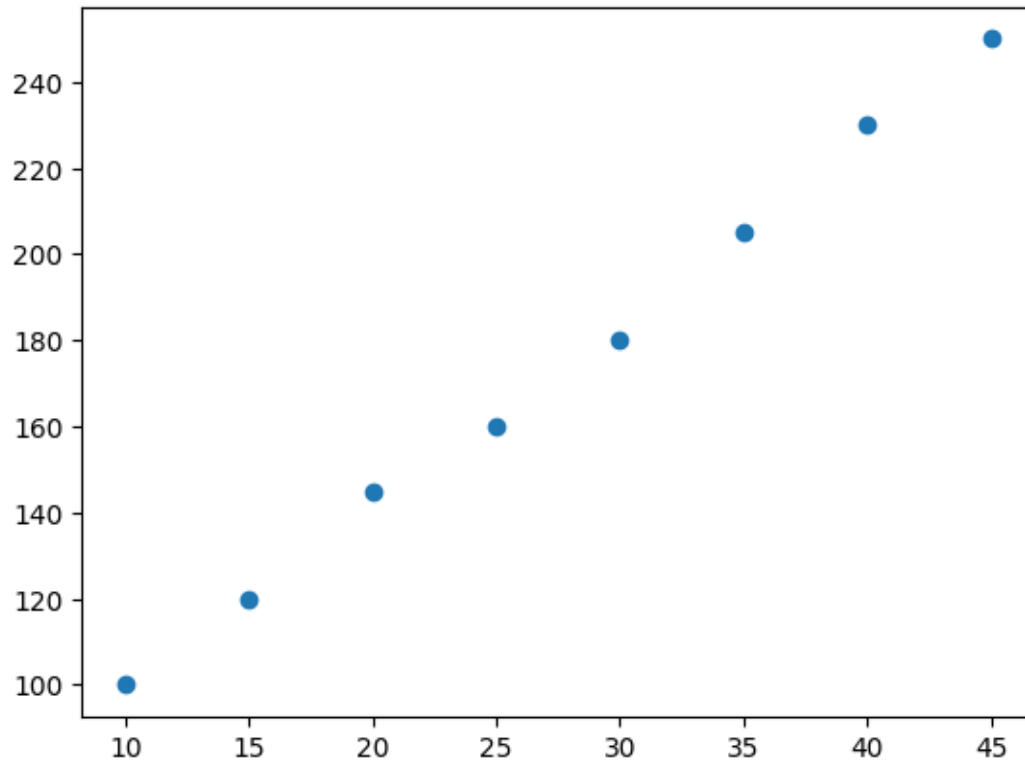


Q3. Scatter Plot – Advertising Expenditure vs Monthly Sales

Python Code:

```
import matplotlib.pyplot as plt
ad=[10,15,20,25,30,35,40,45]
sales=[100,120,145,160,180,205,230,250]
plt.scatter(ad,sales)
plt.show()
```

Output:

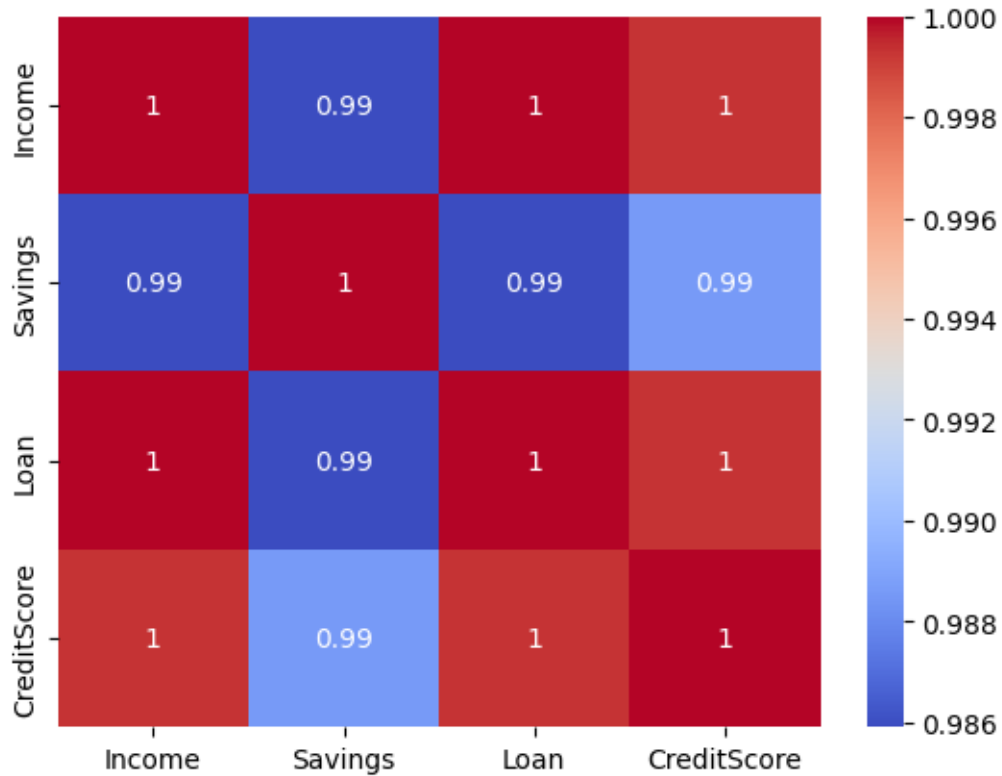


Q4. Correlation Matrix and Correlogram – Financial Institution Dataset

Python Code:

```
import pandas as pd, seaborn as sns, matplotlib.pyplot as plt
df=pd.DataFrame({'Income':[30,40,50,60,70],'Savings':[5,8,12,18,25],'Loan':[15,20,25,30,35],'CreditScore':[620,670,710,760,810]})
sns.heatmap(df.corr(),annot=True,cmap='coolwarm')
plt.show()
```

Output:

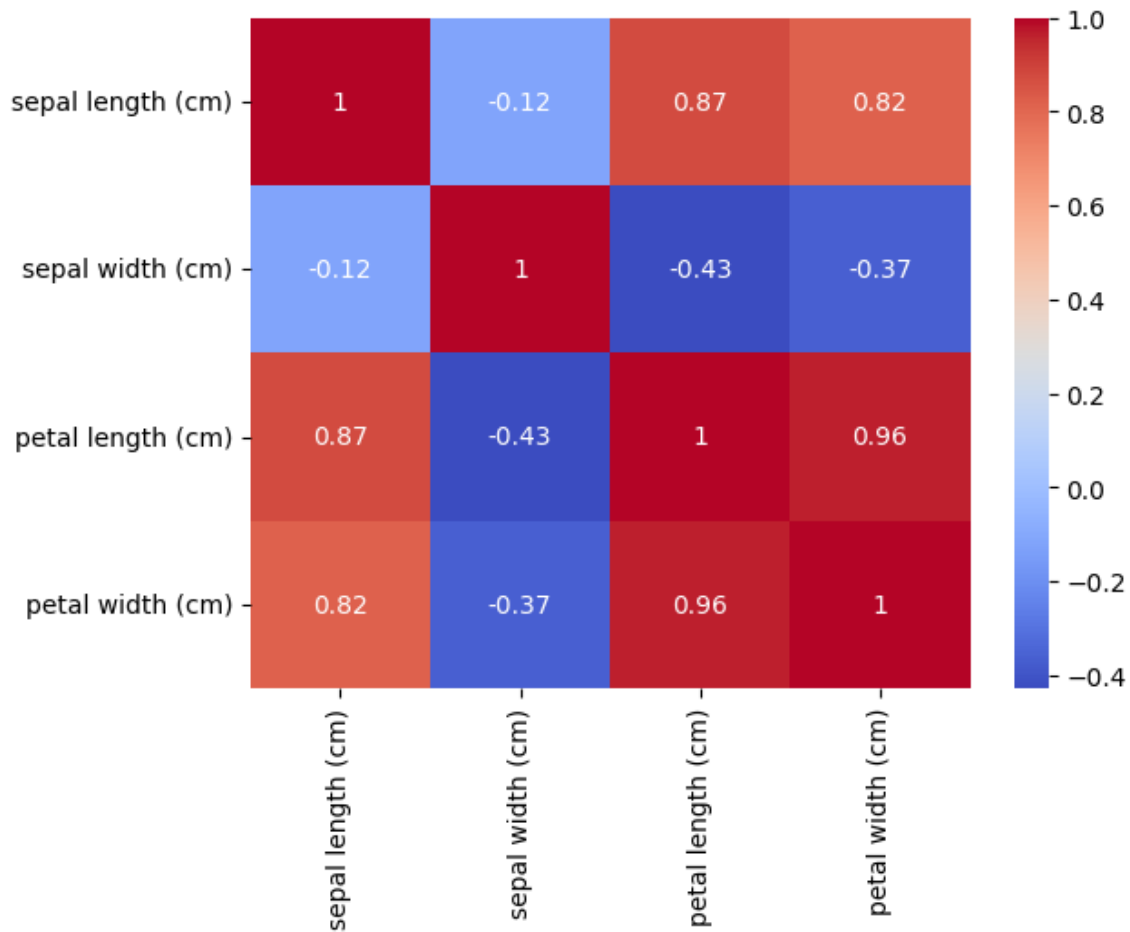


Q5 Correlation Heatmap – Iris Dataset

Python Code:

```
from sklearn.datasets import load_iris
import seaborn as sns, matplotlib.pyplot as plt
iris=load_iris(as_frame=True).frame.iloc[:, :4]
sns.heatmap(iris.corr(), annot=True, cmap='coolwarm')
plt.show()
```

Output:

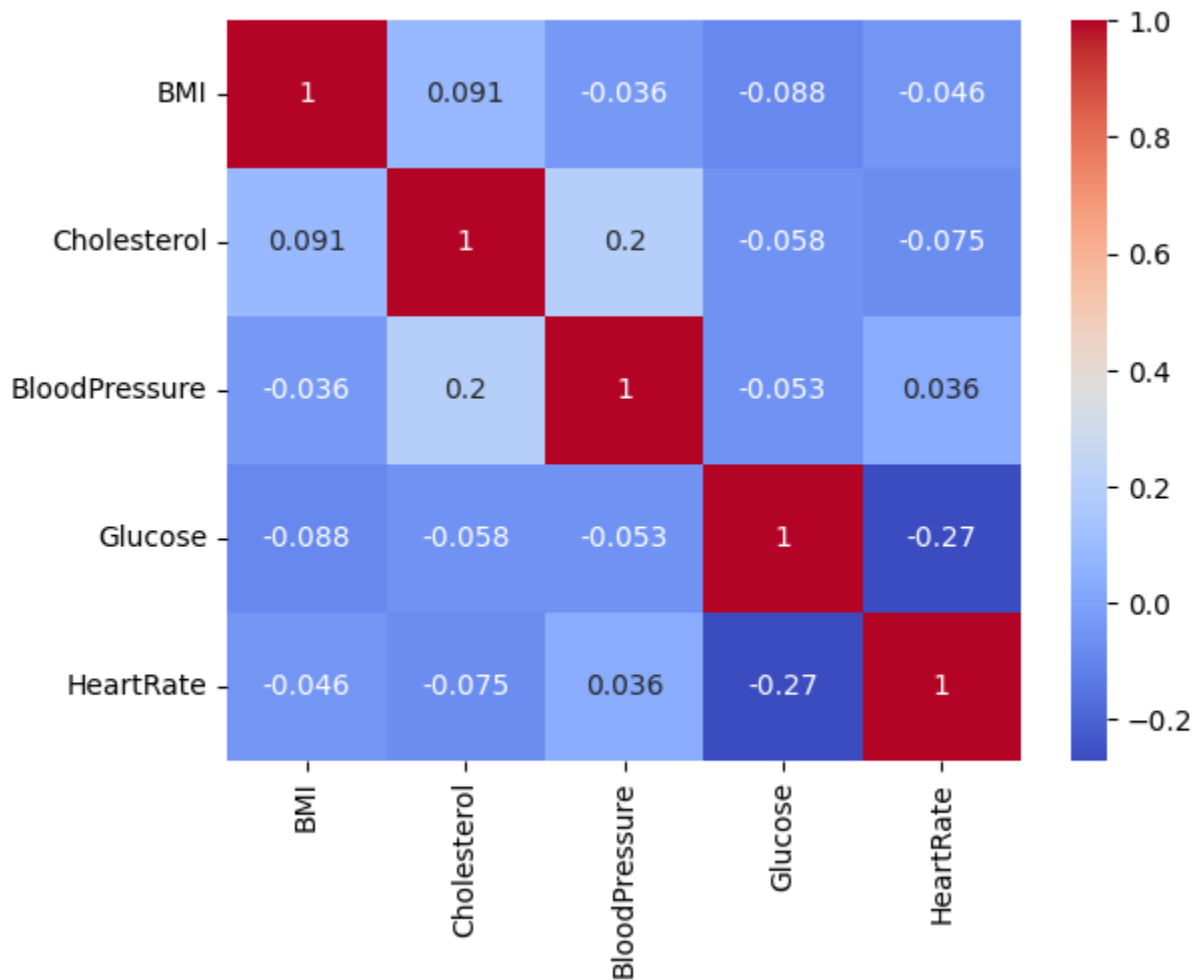


Q6. Correlogram – Healthcare Dataset (Multicollinearity Analysis)

Python Code:

```
import numpy as np,pandas as pd,seaborn as sns,matplotlib.pyplot as plt
np.random.seed(1)
df=pd.DataFrame({'BMI':np.random.normal(27,3,100),'Cholesterol':np.random.normal(200,20,100),'BloodPressure':np.random.normal(120,12,100),'Glucose':np.random.normal(100,15,100),'HeartRate':np.random.normal(75,8,100)})
sns.heatmap(df.corr(),annot=True,cmap='coolwarm')
plt.show()
```

Output:

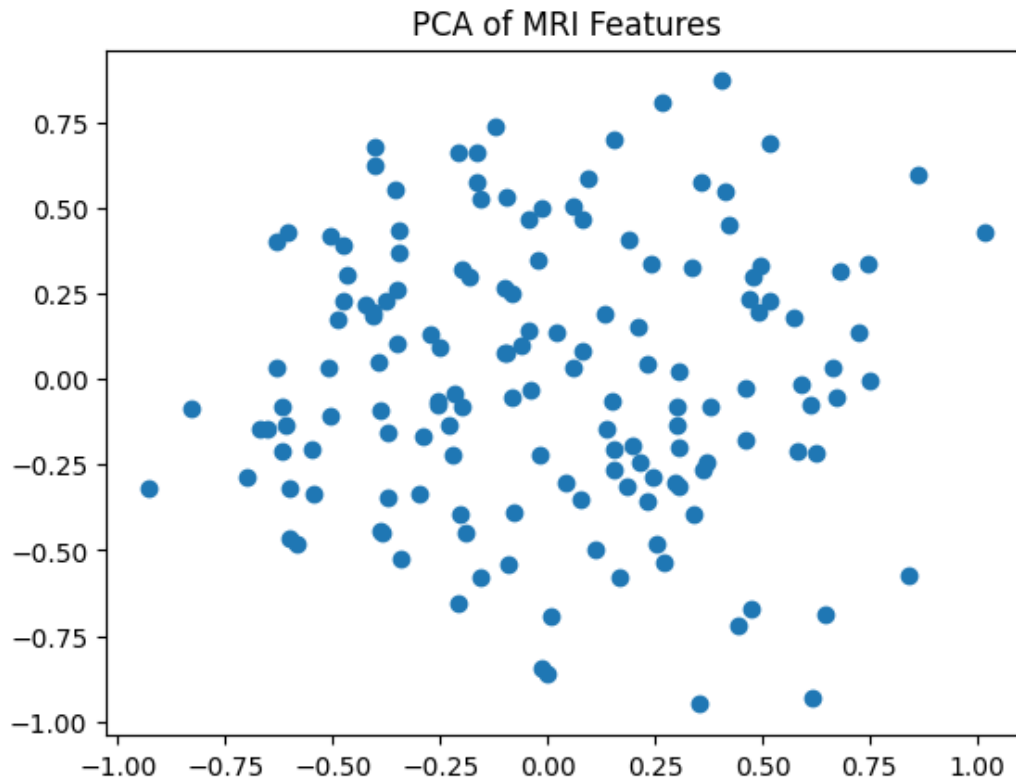


Q7. PCA on MRI Features

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
np.random.seed(42)
X=np.random.rand(150,25)
pca=PCA(n_components=2)
Y=pca.fit_transform(X)
print(pca.explained_variance_ratio_)
plt.scatter(Y[:,0],Y[:,1])
plt.title('PCA of MRI Features')
plt.show()
```

Output:

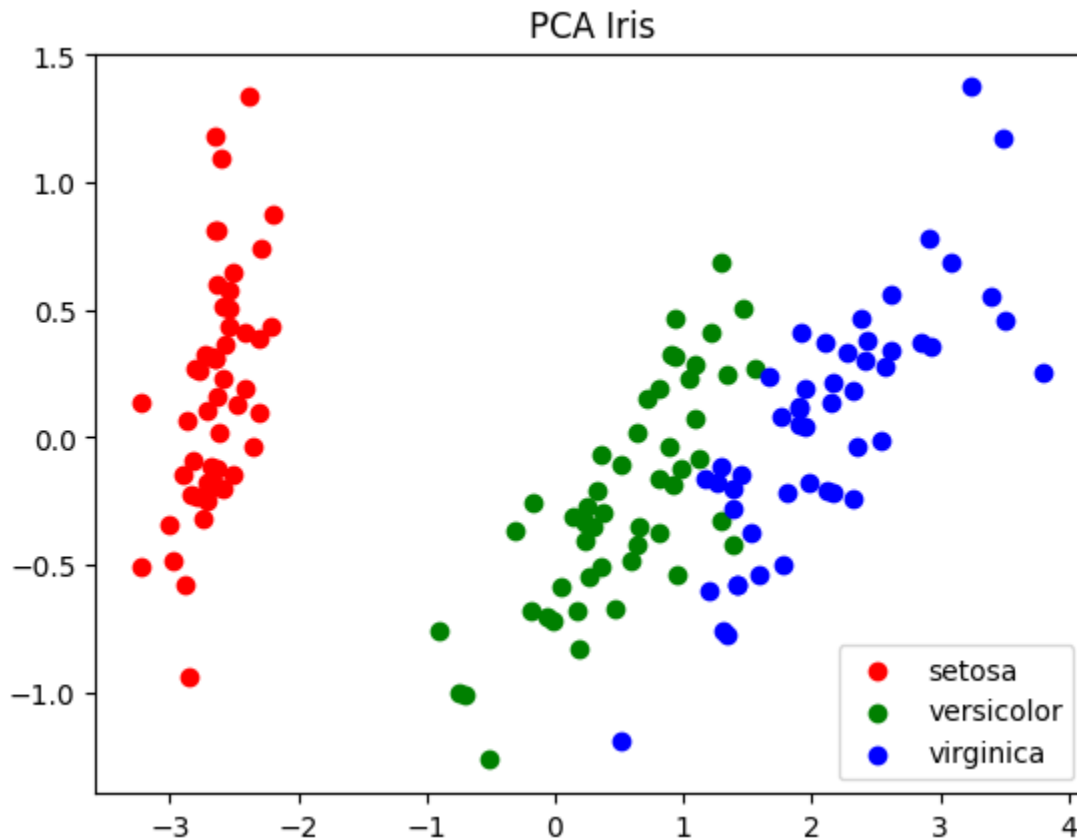


Q8. PCA on Iris Dataset

Python Code:

```
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
iris=load_iris()
pca=PCA()
X=pca.fit_transform(iris.data)
print(pca.explained_variance_ratio_)
for i,c in zip(range(3),['r','g','b']):
    plt.scatter(X[iris.target==i,0],X[iris.target==i,1],c=c,label=iris.target_names[i])
plt.legend();plt.title('PCA Iris');plt.show()
print('PC1 explains the highest variance.')
```

Output:

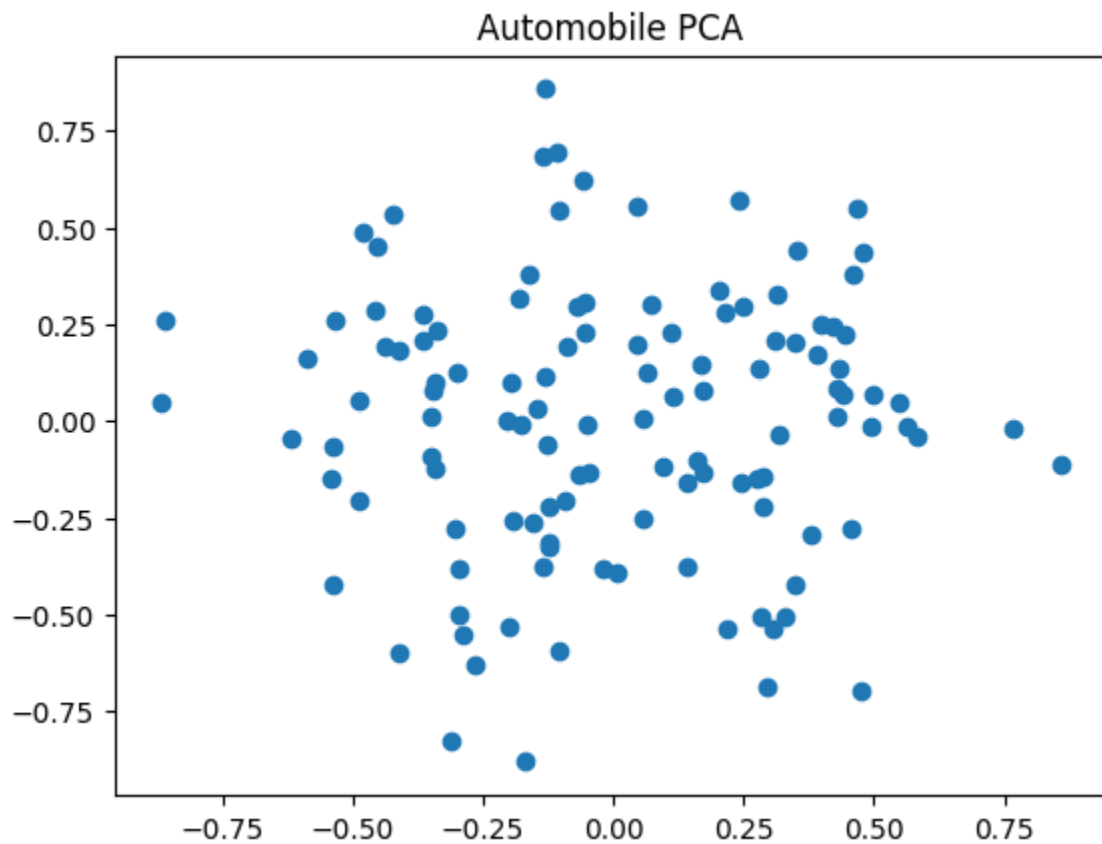


Q9. PCA on Automobile Specifications

Python Code:

```
import numpy as np
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
X=np.random.rand(120,15)
Y=PCA(n_components=2).fit_transform(X)
plt.scatter(Y[:,0],Y[:,1]);plt.title('Automobile PCA');plt.show()
```

Output:

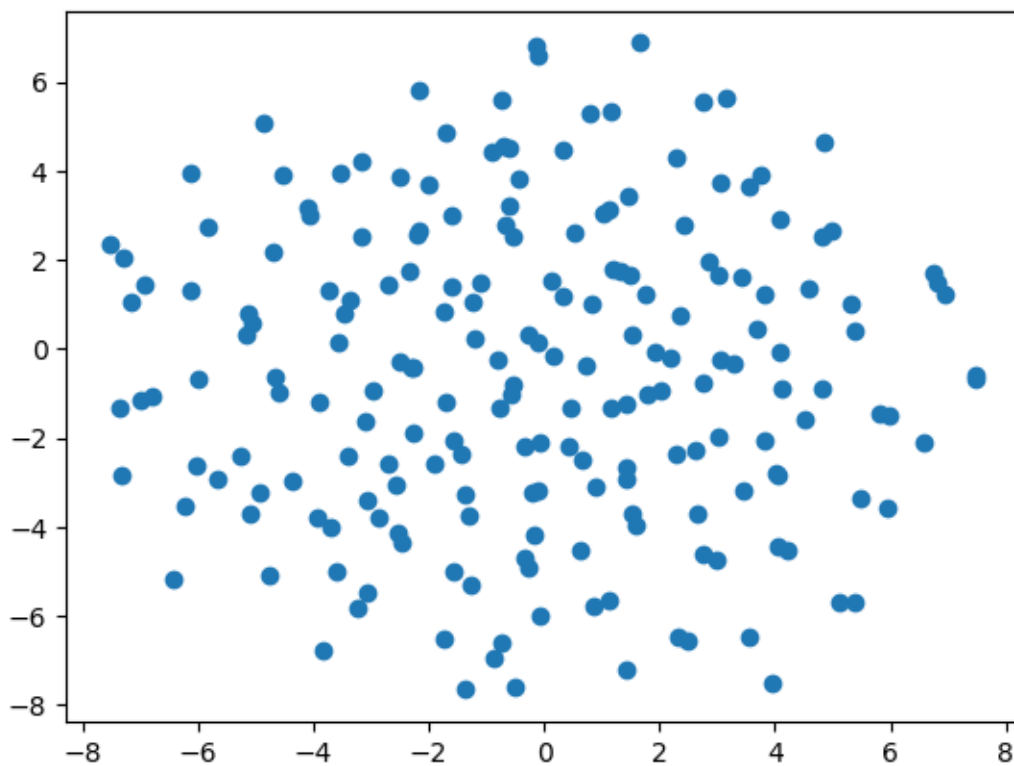


Q10. t-SNE on Facial Embeddings

Python Code:

```
import numpy as np
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt
X=np.random.rand(200,128)
Y=TSNE(n_components=2,random_state=42).fit_transform(X)
plt.scatter(Y[:,0],Y[:,1]);plt.show()
print('t-SNE preserves local structure better than PCA.')
```

Output:

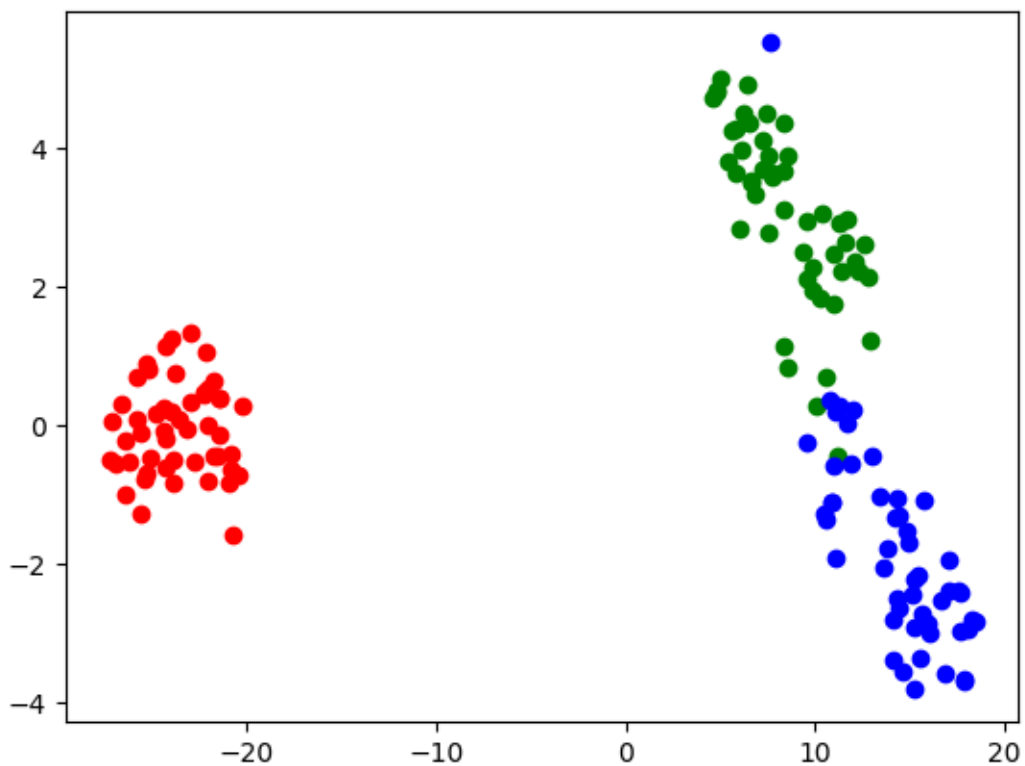


Q11. t-SNE on Iris

Python Code:

```
from sklearn.datasets import load_iris
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt
iris=load_iris()
Y=TSNE(n_components=2,random_state=42).fit_transform(iris.data)
for i,c in zip(range(3),['r','g','b']):
    plt.scatter(Y[iris.target==i,0],Y[iris.target==i,1],c=c)
plt.show()
```

Output:

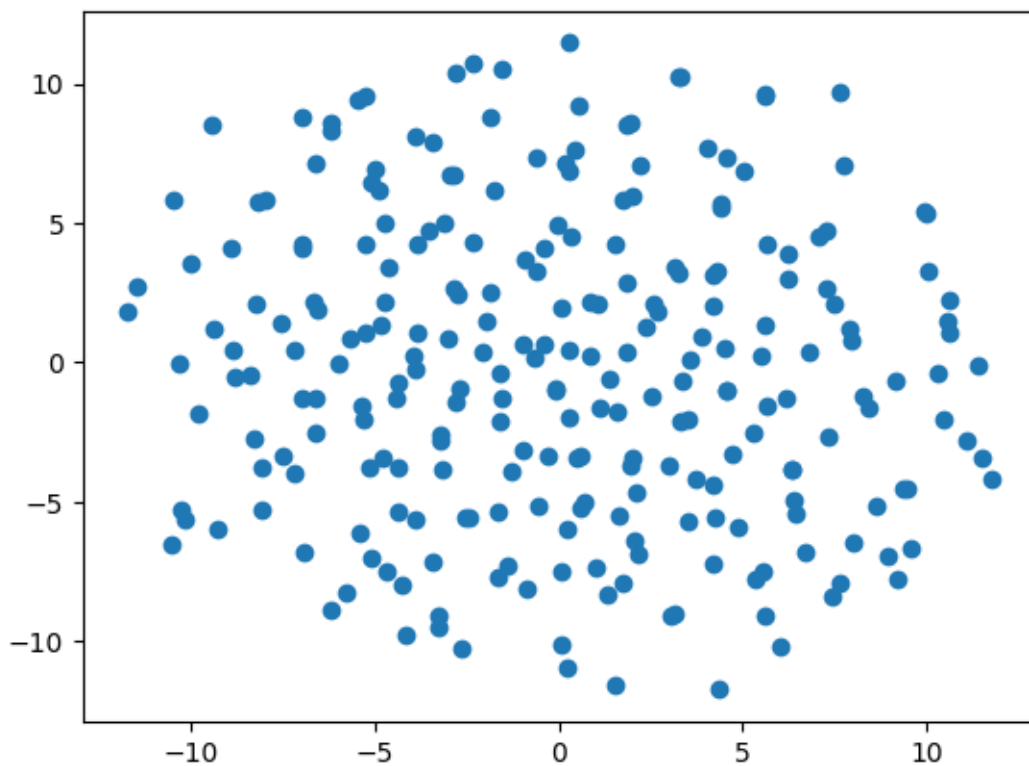


Q12. t-SNE Customer Segmentation

Python Code:

```
import numpy as np
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt
X=np.random.rand(250,40)
Y=TSNE(n_components=2,random_state=42).fit_transform(X)
plt.scatter(Y[:,0],Y[:,1]);plt.show()
print('Check whether distinct clusters appear.')
```

Output:

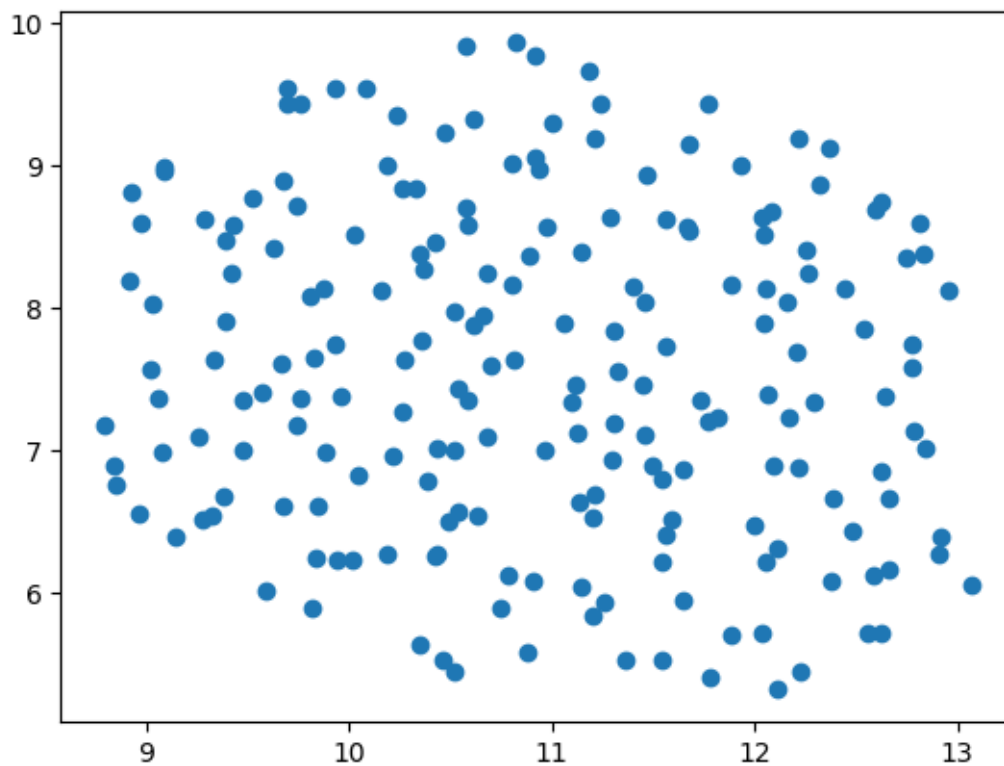


Q13. UMAP Medical Images

Python Code:

```
import numpy as np
import umap
import matplotlib.pyplot as plt
X=np.random.rand(200,100)
Y=umap.UMAP(random_state=42).fit_transform(X)
plt.scatter(Y[:,0],Y[:,1]);plt.show()
print('UMAP usually preserves local structure better than PCA.')
```

Output:

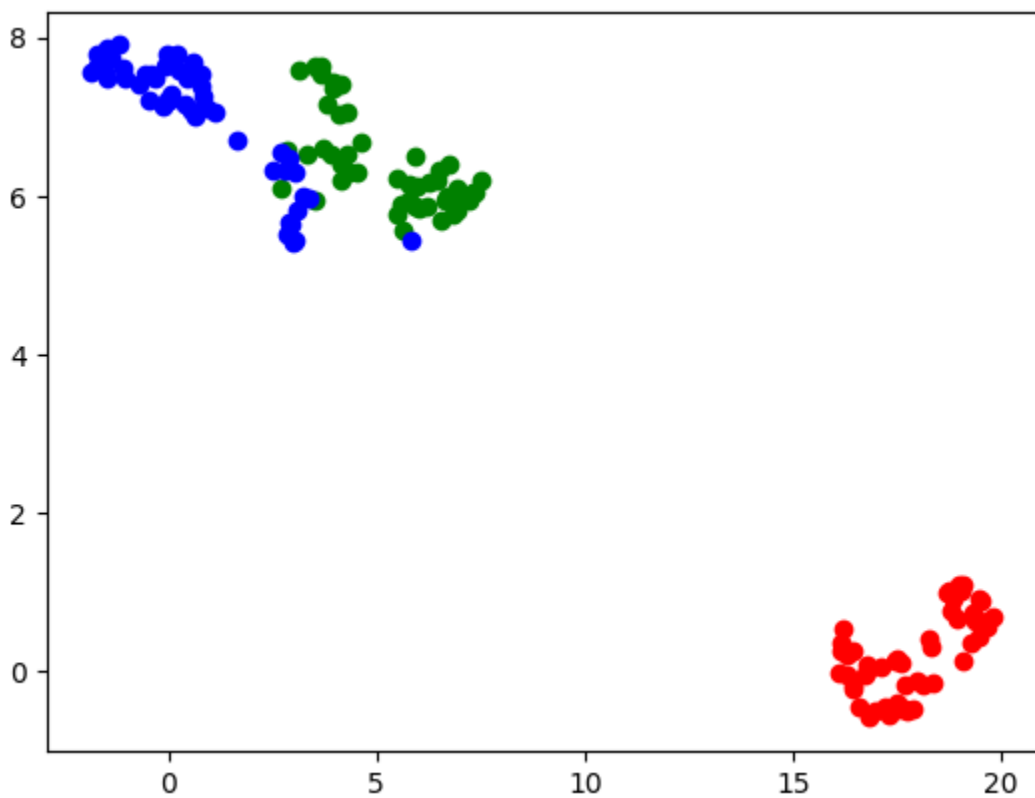


Q14. UMAP on Iris

Python Code:

```
from sklearn.datasets import load_iris
import umap, matplotlib.pyplot as plt
iris=load_iris()
Y=umap.UMAP(random_state=42).fit_transform(iris.data)
for i,c in zip(range(3),['r','g','b']):
    plt.scatter(Y[iris.target==i,0],Y[iris.target==i,1],c=c)
plt.show()
```

Output:



Q15. UMAP E-commerce Data

Python Code:

```
import numpy as np
import umap
import matplotlib.pyplot as plt
X=np.random.rand(300,60)
Y=umap.UMAP(random_state=42).fit_transform(X)
plt.scatter(Y[:,0],Y[:,1]);plt.title('Customer Clusters');plt.show()
```

Output:

